$0.12 \pm 0.03 0.23 \pm 0.10 \times 0.048 0.053 \times 0.147 0.051 \approx 0.000 0.150 \pm 0.019 r = 0.995 \times$

$\times n = 5 \times$

$\times$

$\theta_0 \in \mathbb{R}^d d \theta \in \mathbb{R}^d \ell_2$

$$\Delta W(\theta, \theta_0) = \frac{\|\theta - \theta_0\|_2}{\|\theta_0\|_2} \times 100$$

$AB \theta_A \theta_B$

$$\Delta W(\theta_A, \theta_B) = \frac{\|\theta_A - \theta_B\|_2}{\|\theta_0\|_2} \times 100$$

$\theta_A, \theta_B \in \mathbb{R}^d \alpha \in [0, 1]$

$$\theta(\alpha) = (1 - \alpha) \cdot \theta_A + \alpha \cdot \theta_B$$

$\mathcal{L}(\theta(\alpha))$

$$\boxed{\mathcal{B}(\theta_A, \theta_B) = \underset{\alpha\in[0,1]}{} \mathcal{L}\big(\theta(\alpha)\big) - \frac{\mathcal{L}(\theta_A) + \mathcal{L}(\theta_B)}{2}}$$

$\alpha \in \{0.0, 0.1, \ldots, 1.0\}$
$L\theta^{(\ell)} \subset \theta\ell$

$$\Delta W^{(\ell)} = \frac{\|\theta^{(\ell)} - \theta_0^{(\ell)}\|_2}{\|\theta_0^{(\ell)}\|_2} \times 100, \quad \ell = 0, 1, \ldots, L-1$$

r

$$r = \frac{\sum_\ell(\Delta W_A^{(\ell)} - \overline{\Delta W}_A)(\Delta W_B^{(\ell)} - \overline{\Delta W}_B)}{\sqrt{\sum_\ell(\Delta W_A^{(\ell)} - \overline{\Delta W}_A)^2}\sqrt{\sum_\ell(\Delta W_B^{(\ell)} - \overline{\Delta W}_B)^2}}$$

$$\theta = \theta_0 + \sigma \cdot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, {}_d)$$

$\sigma\Delta W(\theta, \theta_0)\mathcal{B}(\theta_0, \theta)$
$\mathcal{S} \subseteq \{0, \ldots, L-1\}$

$$\theta_{\mathcal{S}}^{(\ell)}(\alpha) = \begin{cases} (1-\alpha)\theta_A^{(\ell)} + \alpha\theta_B^{(\ell)} & \ell \in \mathcal{S} \\ \theta_A^{(\ell)} & \ell \notin \mathcal{S} \end{cases}$$

$\mathcal{B}_{\mathcal{S}} = \mathcal{B}(\theta_{\mathcal{S}}(0), \theta_{\mathcal{S}}(1))\mathcal{S}$
$\theta^{(t)}t = 40, 80, \ldots, 400$

$$\mathcal{B}(t) = \mathcal{B}\big(\theta_0,\ \theta^{(t)}\big)$$

$\Delta W \to \mathcal{B}$
td

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}, \qquad s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}}$$

$|d| > 0.8|d| > 0.5|d| > 0.2$
$\Delta W \approx 1.4\Delta W \approx 8$

$r = 0.995\sim$

$\Delta W \approx 1.4\sim$

$\Delta W \approx 8\times$

$\times\times$

$\Delta W \approx 5\Delta W \approx 4.6\times$

$\sim$

$\beta_1 = 0.9, \beta_2 = 0.999$

	$\pm\sigma$	
ΔW	ΔW	$\leftrightarrow$
1.4 ± 0.0	1.5 ± 0.1	2.0 ± 0.1
8.0 ± 0.3	8.5 ± 0.2	11.6 ± 0.3

$\approx \leftrightarrow \leftrightarrow$

$n = 5 \binom{5}{2} = 10$

$\Delta W \Delta W \approx 8$

$\Delta = -0.071 \Delta = +0.161$

0.150 ± 0.019

$\Delta W \rightarrow \Delta W \rightarrow \mathcal{B}$

$\Delta W \approx 0.3 \Delta W \approx 1.1 \Delta W \approx 1.4$

$\Delta W \in \{0.5, 1, 2, 4, 8\}$

$\Delta W = 8 \approx \Delta W = 1.5\times$

$r = 0.995\times$

td

$p = 0.55 d = 0.49 p < 0.001 d = 3.66 p < 0.001 d = 4.17 p < 0.001 d = 5.27 |d| > 0.8$

$\bar{\boldsymbol{\theta}} = (\boldsymbol{\theta}_A + \boldsymbol{\theta}_B)/2 \mathcal{L}$

$$\mathcal{B}(\boldsymbol{\theta}_A, \boldsymbol{\theta}_B) \approx \frac{1}{8} \Delta\boldsymbol{\theta}^{\mathsf{T}} \, (\bar{\boldsymbol{\theta}}) \, \Delta\boldsymbol{\theta}$$

$\Delta\boldsymbol{\theta} = \boldsymbol{\theta}_B - \boldsymbol{\theta}_A \|\Delta\boldsymbol{\theta}\|$

$\rho = (\Delta\boldsymbol{\theta}^{\mathsf{T}} \Delta\boldsymbol{\theta} / \|\Delta\boldsymbol{\theta}\|^2) / (()/d) \rho = 1 \rho \gg 1 \times \Delta W \rho / \rho \approx 9$

$\propto e^{-\delta\ell} (_\ell) \propto e^{-\gamma\ell} \mathcal{B}_\ell \propto e^{-(\delta+\gamma)\ell} \delta + \gamma \approx 0.080.12$

$r = 0.995$

$d\boldsymbol{\theta} = -\eta \nabla\mathcal{L} \, dt + \sqrt{\eta \boldsymbol{\Sigma}} \, d\boldsymbol{\Sigma} = \|\nabla\mathcal{L}\|^2 / \, (\boldsymbol{\Sigma})$

$\rightarrow \boldsymbol{\Sigma} \rightarrow \rightarrow \rightarrow \rightarrow \boldsymbol{\Sigma} \rightarrow \rightarrow \rightarrow$

$\pm\sigma$

0.053 ± 0.011	0.051 ± 0.013
0.118 ± 0.031	0.228 ± 0.102

0.048 ± 0.000	
0.071 ± 0.007	
0.087 ± 0.032	
0.147 ± 0.027	$\times$

$$P() \approx \left(-\Delta\boldsymbol{\theta}^{\mathsf{T}}\Delta\boldsymbol{\theta}/(2\kappa)\right)$$

$\kappa\kappa{\rightarrow}\kappa{\rightarrow}\times$

$\boldsymbol{\Sigma}$

$\lambda > \lambda$

$\Delta W \rightarrow \mathcal{B}$

$\Delta W = 8\Delta W = 1.5{\times}\Delta W$

$+0.061{-}0.058{\times}{\times}$

ρ

$\times\Delta W$

$r = 0.995$

$\leftrightarrow$

$\Delta W \approx 0.3{\approx}\ 0.001\Delta W \rightarrow \mathcal{B}{\sim}$

$n = 3\sigma^2 < 0.001 n = 5 n = 3 n = 10{\times}n = 3$

$n = 5$

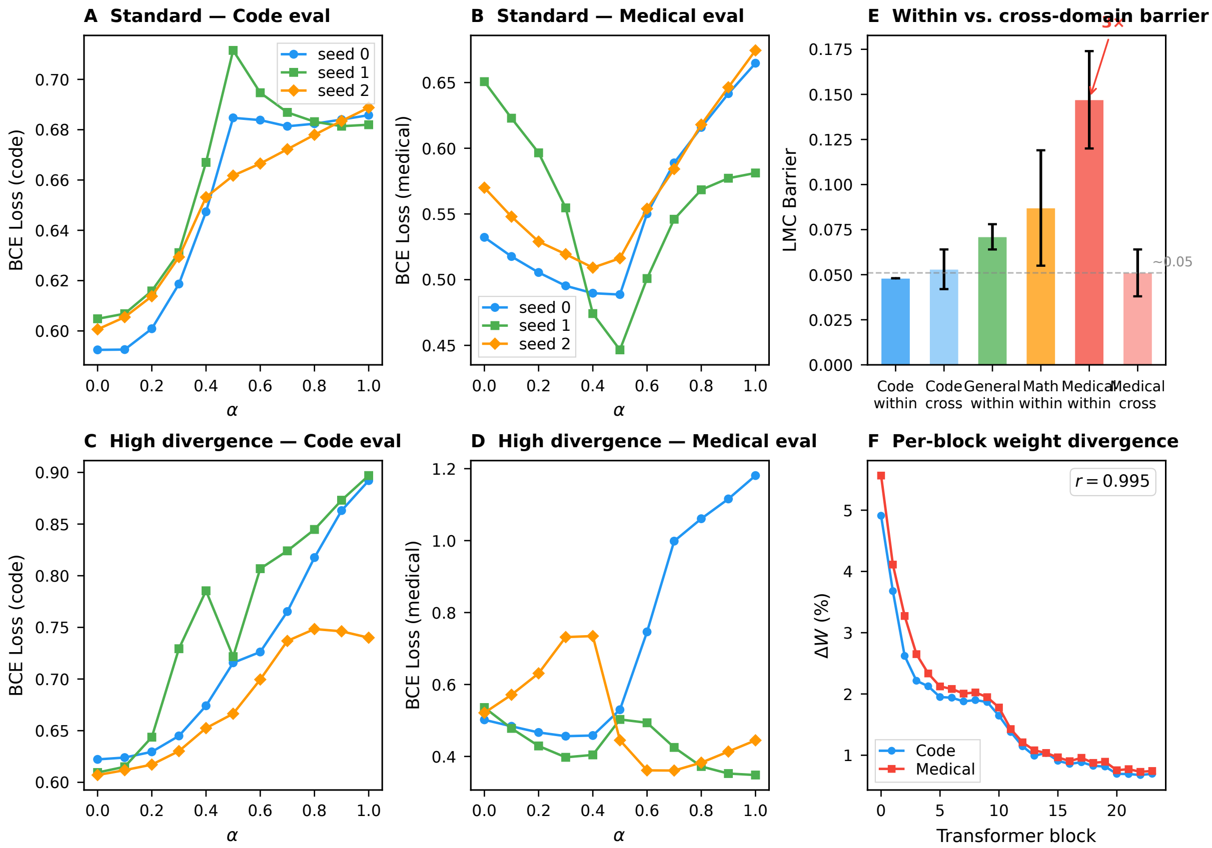
$\leftrightarrow$	
$\leftrightarrow$	
$\leftrightarrow$	
$\leftrightarrow$	
$\leftrightarrow$	
$\leftrightarrow$	
$\leftrightarrow$	
$\leftrightarrow$	
$\leftrightarrow$	
$\leftrightarrow$	
$\pm\sigma$	0.304 ± 0.322
$\pm\sigma\leftrightarrow$	0.203 ± 0.116

0.599 ± 0.005	0.584 ± 0.049
0.686 ± 0.003	0.663 ± 0.007

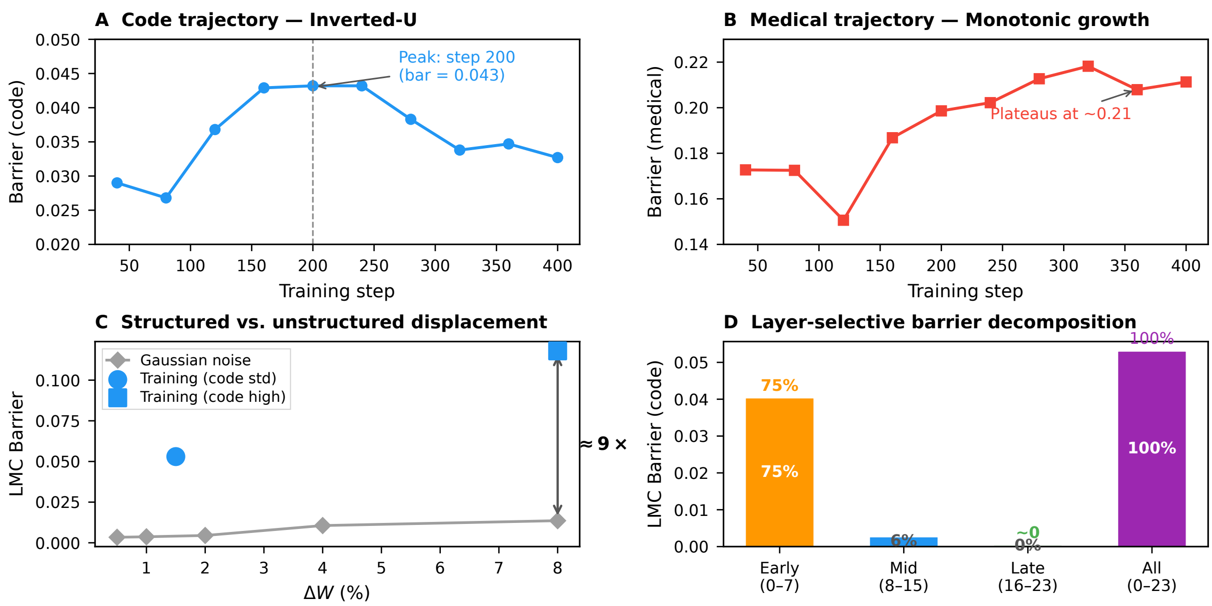
	≈ 0.000	≈ 0.000
$\leftrightarrow$	0.033 ± 0.001	0.150 ± 0.019

ΔW

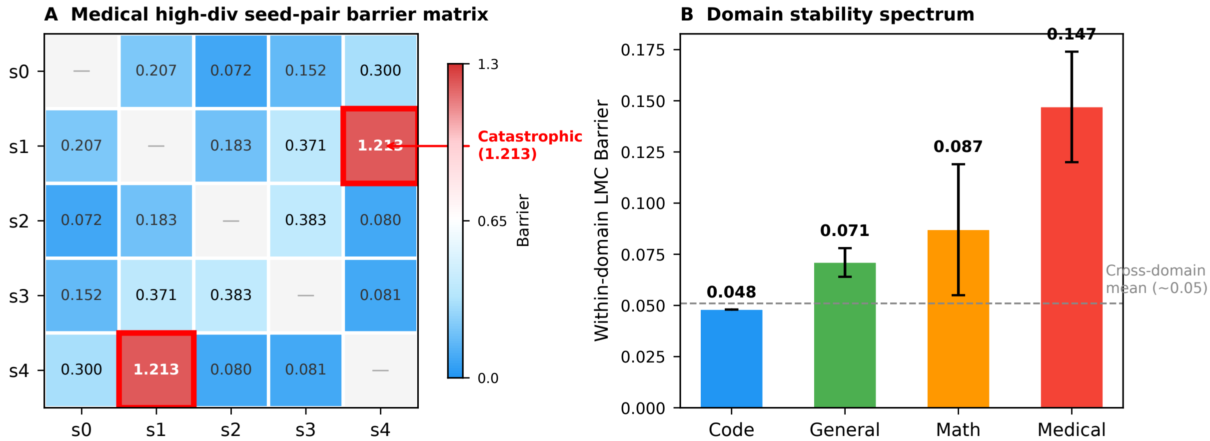
	t	p	d
$\approx$	$t(2.0) = 0.60$		
$>$	$t(2.9) = 4.49$	$<$	
$\neq$	$t(2.0) = 5.11$	$<$	
$>$	$t(4.0) = 3.44$	$<$	
$>$	$t(2.5) = 2.83$		
$>$	$t(2.1) = 2.45$		



$$\theta(\alpha) = (1 - \alpha)\theta + \alpha\theta \times \pm \sigma r = 0.995$$



$$\Delta W \sim \Delta W \sim \times$$



$$\binom{5}{2} = 10 \times \leftrightarrow \sim$$